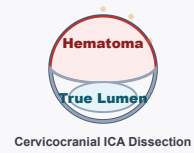
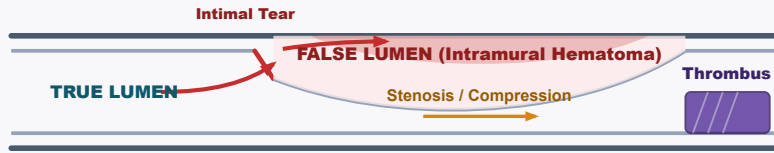
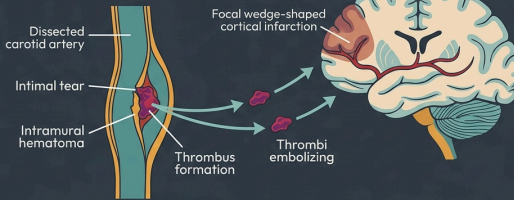


Cervicocranial Artery Dissection

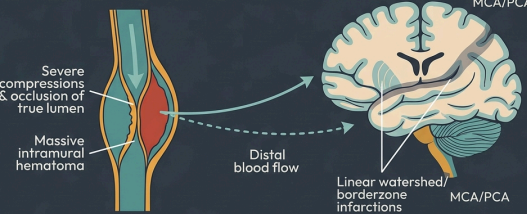


Cervical Artery Dissection (CeAD)

Part 1: Thromboembolism



Part 2: Hemodynamic Hypoperfusion



1. Clinical Presentation & Pathophysiology

Ipsilateral Pain & Onset

- **Carotid (ICA):** Frontotemporal/retro-orbital/facial pain (jaw angle).
- **Vertebral (VA):** Severe occipital or posterior neck pain.
- **Onset:** Precedes stroke/TIA by hours to days (median 4 days).

Anhidrosis-Sparing Horner's

- **Signs:** Ptosis/miosis (28–58% of ICA) **without** anhidrosis.
- **Mechanism:** Sweat fibers follow ECA plexus; pupil/eyelid fibers follow ICA.

Neurological Deficits

- **CN Palsies:** CN IX–XII palsies (8–16%) from local ICA compression.
- **VA Territory:** Wallenberg syndrome, cerebellar ataxia, PICA/AICA strokes.

2. Diagnostic Workup

- **CTA Head/Neck:** Shows string sign, dissection flap, pseudoaneurysm, or occlusion.
- **MRI Neck (T1 Fat-Sat):** Pathognomonic crescent sign (intramural hematoma).
- **DSA:** Reserve for diagnostic doubt or stenting.
- **Screening:** Assess for FMD/connective tissue disease, especially if spontaneous/recurrent.

3. Medical Management: Extracranial vs. Intracranial Dissection

Extracranial Dissection

- **Antithrombotics:** Continue for at least 3–6 months (Class I, ESO/AHA).
- **Choice:** SAPT vs VKA/DOAC; short-term DAPT (21–90d) is reasonable (ESO consensus).
- **STOP-CAD:** In occlusions, consider anticoagulation Day 1–30, then switch to antiplatelet.
- **IV Thrombolysis:** Safe & indicated within 4.5 hours (Class I).

Intracranial & Pseudoaneurysms

- **SAH:** Lack external elastic lamina & thin adventitia; rupture risk.
- **Anticoagulation:** Avoided if SAH present. Prefer single antiplatelet.
- **IVT Caution:** IVT is safe in extracranial CCAD but safety/efficacy in intracranial extension is not well established (AHA 2024).
- **Stenting:** Reserve for recurrent ischemia despite optimal medical therapy or severe flow-limiting stenosis.

- **Recurrence & Activity:** Long-term CCAD recurrence is low (~1%/yr). Avoid high-risk neck activities (chiropractic neck manipulation, rollercoasters, extreme hyperextension/rotation) for secondary prevention.

4. Landmark Trial & Cohort Evidence

Study / Year	Population & Design	Interventions Compared	Key Outcomes & Clinical Nuance
CADISS 2015	N = 250. Extracranial CCAD. RCT.	Antiplatelet vs. Anticoagulant for 3 months.	• Primary Composite (Ipsilateral stroke or death at 3m): 2.0% vs. 1.0% (p = 0.63). Established clinical equipoise.
TREAT-CAD 2021	N = 194 (PP = 173). Extracranial. RCT.	Aspirin 300mg daily vs. VKA for 3 months.	• Primary Composite (Stroke, major hemorrhage, death, or new MRI lesion at 14d): 23% vs. 15% (Non-inferiority NOT met). Ischemic stroke: 8.0% vs. 0%.
Kaufmann IPD 2024	N = 444. IPD meta-analysis of CADISS + Antiplatelet vs. Anticoagulant x 90d. TREAT-CAD.	Antiplatelet vs. Anticoagulation.	• Primary Composite: No diff (1.4% anticoagulation vs. 4.4% antiplatelet, p = 0.11). • Ischemic Stroke alone: Significant reduction with anticoagulation (0.5% vs. 4.0%; OR 0.14, p = 0.01), with a non-significant increase in major bleeding (0.9% vs. 0%).
STOP-CAD 2024	N = 3,636. Multicenter observational cohort registry.	Antiplatelet vs. Anticoagulation.	• Temporal Risk: 87% of recurrent strokes occurred in the first 30 days. • Occlusion Benefit: Patients with complete arterial occlusion at baseline had the highest stroke risk and benefited most from early anticoagulation. • Transition Strategy: Initiate anticoagulation for days 1–30 (highest stroke risk window) and then transition to antiplatelet monotherapy at day 30 to mitigate long-term bleeding risks (which rise significantly by day 180).

CADISS: *Lancet Neurol.* 2015;14(4):361-7. PMID: 25684164 | TREAT-CAD: *Lancet Neurol.* 2021;20(5):341-350. PMID: 33765420
Kaufmann IPD: *JAMA Neurol.* 2024;81(6):630-637. PMID: 38739383 | STOP-CAD: *Stroke.* 2024;55(4):908-918. PMID: 38334460 | AHA/ASA: *Stroke.* 2021;52:e364-e467. PMID: 34024117 | AHA Statement 2024: *Stroke.* 2024;55:e84-e107. PMID: 38301552 | ESO Guideline 2021: *Eur Stroke J.* 2021;6(3):XXXIX-LXXXVIII. PMID: 34528453